

Assignment 9 Library Management System using React, NodeJS and MySQL

Deadline: 22/03/26

Build a Library Management System using React for the frontend, NodeJS/Express for the backend, and MySQL for the database. Implement the following functionalities:

- Design the React components for Home, AddBook, EditBook and DeleteBook pages.
- Create RESTful API endpoints in Node.js for **CRUD operations** with MySQL.
- Ensure **proper routing and navigation** after Add, Edit, and Delete actions.
- Handle errors and display appropriate messages to the user.

a) The homepage should look as follows listing all the books from the database:

Library Management System

[Home](#) [Add Book](#)

Book List

ID	Title	Author	Year	Actions
1	React Basics	John Doe	2022	<button>Edit</button> <button>Delete</button>
2	Node.js Guide	Alice Smith	2023	<button>Edit</button> <button>Delete</button>
3	MySQL Handbook	David Lee	2021	<button>Edit</button> <button>Delete</button>

b) Edit button when clicked should allow editing the book details and the interface should look as follows.

Library Management System

[Home](#) [Add Book](#)

Edit Book

Book updated successfully!

Title

Author

Year

Note: After editing redirect the user to the Home page where the new book appears in the list without a full page reload using **React state management**.

c) Delete button when clicked should allow editing the book details and the interface should look as follows.

Library Management System

[Home](#) [Add Book](#)

Delete Book

Are you sure you want to delete this book?

Title: React Basics

Author: John Doe

Year: 2022

Yes, Delete

Cancel

Note: After deleting redirect the user to the Home page where the updated book list appears without a full page reload using **React state management**.